

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper B)

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

- The business interruption insurance can cover for lost revenue due to not being able to conduct business due to property damage etc.
 - The natural disasters seem to have increased in the frequency and it may be necessary to have business interruption cover.
 - In addition, there could be disruption on account of pandemics due to government mandated lockdowns.
 - Therefore, it ensures that a temporary shutdown in the plant won't result in indefinite closure
 - Part of the claim amount can be used to repair the damage to the property and machinery caused
 - One of the purposes of business insurance is to allow to pay the workers their salaries and wages
 - This ensures that manufacturer retains employees, even if the plant is temporarily closed.
 - Having to re-hire staff will create a handicap upon reopening, as one will have to train the new employees.
 - Also, it may be costly to rehire new staff
 - If the manufacturer suffers damage the plant, it may still have to pay any leases and therefore portion of the claim amount can be used to cover rental expenses.
 - In case it may be necessary to shift the plant then the insurance may cover that as well
 - If business loan is taken, then part of the claim amount can be used to pay the loan re-payment.
 - This will also ensure that any interest-on loan is not accumulated during the closure of business
- [1/2 mark each, **Max 4**]

ii)

- Ownership (e.g., of land, buildings, equipment, and inventories)
- The age and design of the plant and the manufacturing facilities
- The plant location and the nature of activities performed at the site
- Any previous hazardous events in the area impacting business
- Any previous history of business interruptions and losses on account of the same.
- Neighbourhood hazards like close to chemical plant
- Previous and current insurance covers
- Financials: Annual sales, profits, expenses, or revenue generated by the plant or business
- Organisational structure of the plant, including decision-making and production structures
- Substitution possibilities for the individual product/business areas because of a business interruption
- Example alternative production, relocation to other plants, contract manufacturing
- The business continuity management and safety measures employed at the plant
- The frequency of audit and the up-dation of the business continuity and safety practices
- Willingness of the policyholder to assess risks and implement risk management procedures
- Necessary certifications, licenses, approvals, etc.
- For example, fire safe certificate issued by respective government department from time to time
- Susceptibility of the production plant (e.g., just-time-delivery of raw materials)
- As well as the goods produced/stored (e.g., to fire, heat, smoke, humidity).
- Estimated time needed for recovery of the operations
- Required reconstruction times for destroyed/damaged buildings including the procurement of the necessary permits
- As well as any expected reconstruction restrictions due to official requirements
- Indication of the terms for delivery/replacement and installation times required for important machines and plants (series machines vs. custom-made vs. self-built)
- Time required to regain important operating licences and licences to resume production for operations that require permission from authorities
- Stockpiling of important spare parts and reserve systems

- Number of operating shifts in the course of a year
- Degree of utilization of the plant in the course of a year
- Seasonal fluctuations in the individual product and business segments
- Number of permanent employees vs. percentage of employees with contracts for work and services
- Supplier structure of the manufacturer, including information on the proportion of most important suppliers and possible alternatives

[1/2 mark each, **Max 13**]

iii)

- The poor mortality experience of the scheme during the last year
- This means post application of the experience rating the premium rate would increase
- The high claim experience could be on account of
 - Cat event or major accident in the plant
 - Random fluctuations
- If the scheme is reinsured, the reinsurer might have increased the rates
- Due to poor experience during the previous year for similar portfolio of schemes
- Or hardening of reinsurance rates in general due to reduced reinsurance capacity
- The insurer has revised the premiums under the plan which can lead to increase in premium rate due to
 - Higher allowance for expenses
 - Change in any taxation impacting pricing
 - Increase in the cost of capital due to regulatory changes or rate of return requirements on capital
- The age of the scheme has increased significantly over the past year
- This could be on account of employees leaving and new employees of higher age joining
- Change in the risk profile of the scheme due to
 - The higher proportion of risky occupation as compared to last year like machine operators
 - Or relative increase in the sum assured at higher ages
 - Or relative increase in the sum assured for risky occupation like plant operators
- The insurer might be writing the business on loss basis to shore up the top-line earlier
- However, this might have changed due to heavy losses on the business during last year
- Also, such business consumes heavy capital so may not be interested any longer in writing business leading to proposing increase in premium rates
- The lack of competition in the market to provide quotations

[Max 7]

iv)

- The proposal will make sense if the CFO strongly expects the payout to be lower than the premium
- The insurer's premium is likely to contain margin for profit, expenses etc.
- However, his estimation is likely to be crude without any proper data or means of estimation as compared to the insurer's
- The number of deaths at the manufacturer could be less predictable because of risky nature of group
- This means that the variation in the claims could be high making self-insurance a risky proposition
- This will also depend on the level of the sum assured offered to the members

- If the sum assured is low, then it may make sense to take the risks and not insure
- This will also depend on the risk appetite of the manufacturer
- If self-insuring restricts the ability to take risk or expand business, then there will be indirect costs
- If possible, such indirect costs need to be evaluated as part of the proposal
- It may also lead to liquidity issue for the manufacturer if the claims are to be paid at inappropriate time
- The higher claims also will lead to adverse impact on profit
- In such a case it may divert attention of the management which will impact business
- It could explore purchasing the plan for high-risk cases and can self-insure rest of the employees
- This would help it to reduce the premium amount to be paid to insurer and also reduce the risk

[1/2 mark each, **Max 6**]

v)

- Counterparty failure

The default could occur on the investment made by the insurer like default on corporate bonds, deposits etc.

In addition, default could occur on the various contractual obligations of the insurer

- Reinsurance recoveries
- Derivatives contracts

- Inability to meet new business premium

Where new business flows are used to pay current claims, while keeping past premiums invested in less liquid assets, a liquidity risk occurs if new business flows is less than expected.

Premium collection may fail due to a problem with internal systems or external payment systems. While the inability to collect premiums will impact on liquidity to meet short term claims and expenses, it is unlikely that this would last long-term.

- Catastrophe Risk

Unexpected catastrophic exposure can have additional drain on liquidity particularly on concentrated portfolio like Group Term Life plans. This will also depend on the catastrophe reinsurance coverage and delay in settlement by the reinsurer.

- Margin calls

Margin calls under derivative contracts will impact cash resources or assets that can be pledged. Such calls may arise as a result of e.g. a credit rating downgrade, requiring collateral to be posted.

- Mass Surrenders

Probably the key liquidity risk faced is the loss of confidence in the insurer leading to mass surrender claims. The impact of a bank failure/ NBFC failure might also lead to loss of confidence in the financial system in general and possibly mass surrenders.

Looking to the future, greater online processing may increase the number of surrenders that can be transacted in a given period and this will exacerbate the strain in this scenario.

- Risk Management liquidity

In the event of a deterioration of the risk exposure or breach of the risk limits there might be need to off load part of the assets particularly during an unfavorable market condition.

[1 mark each, **Max 6**]

vi) Strategy:

- Move the assets to liquid funds or investments
- Take strategic call to partly move the investment to the sectors which are potentially under valued
- Sell equity futures to lock into a fixed price for the investment.

- Any fall in the value of the investments is then offset by profits on the futures.
- Buy put options
- by buying put options on the assets, the minimum value of the holding can be fixed at the exercise price of the options.
- If the value of the asset goes up the potential profit has only been reduced by the amount of the option premium.

Risks:

- If the investments are liquidated, then if markets rise the investor misses out on the upside.
- Investment in cash may cause asset liability mis-match
- Tax liabilities
- profit is likely to be made due to high equity valuations
- Realizing gains may cause volatility in the P&L
- May not be able to buy derivatives/options for all the investments in the portfolio
- Basis risk as a result of purchasing the derivatives
- For all derivative-based strategies, counterparty/default risk is a further issue when OTC approaches are used.
- Liquidity issue due to margin requirements
- Legal /operational risks due to the complexity of the arrangements.

[1/2 mark each, **Max 7**]

vii)

- The impact on solvency both current and projected post payment of the expected dividend
- The future growth and expansion plans will put restriction on amount of the dividend that can be given
- The Board is also likely to have set a risk appetite that maintains a buffer over the regulatory level of solvency which will again restrict the amount of dividend.
- Any regulatory restriction placed on payment of the dividend even if the solvency is not threatened
- The insurer will also need to consider the dividend it paid in previous years
- its target dividend ratio
- As most companies want to maintain a smooth dividend and want to avoid drop in dividends
- The tax position needs to be considered as it needs to be optimised
- Any potential change in solvency regulation which could impact the solvency capital requirement
- Shareholder expectations need to be taken into consideration
- Example they may prefer share buybacks as opposed to paying the dividends
- And the extent to which investors are influenced by dividend policy within the financial markets
- The assets part of the surplus may be illiquid
- If the prevailing economic conditions are adverse, it may force company to retain the profits
- Prevailing non-economic conditions like probability of higher claims from covid etc also need to be considered
- Any restrictions placed on the level of the dividend by existing bondholders to make sure they get paid.

[1/2 mark each, **Max 7**]

[50 Marks]

Solution 2:

- i) The benefits received by the employee post-retirement though defined in advance may be less valuable or insufficient due to
- The benefit is not linked to inflation or inflation is higher than expected.
 - The circumstances of the employee might have changed making the benefits insufficient.
 - The employer has changed the benefits due to shortfall in fund in case of a funded scheme.
 - The employer has changed the benefits due to higher cost and a downturn in business in the case of a “pay as you go” unfunded scheme.
 - Changes in tax or other regulations resulting in a lower post-tax pension in the hands of employees.
 - Typical merger and acquisition scenario where the sponsor has changed resulting in reduction in promised benefits
 - Insolvency of the sponsor
 - Default in benefit payments or failure to pay benefits in time.
 - Fraud or mismanagement of the funds
 - Inadequate communication by the provider giving rise to complaints and compensations.
 - Any other valid point

The company may not be able to provide the promised benefits due to the following reasons

- Benefits that are promised in the scheme at the outset not viable in the long term.
- Insufficient provisions were made in the case of an unfunded scheme.
- Inappropriate assumptions used for calculation of premiums.
- Insufficient funds have been set aside in case of a funded scheme
- Default in paying contributions in time in the case of a funded scheme
- The investments have underperformed
- The available assets are illiquid to pay the benefits in time.
- Lack of long-term investment options thereby affecting the return on funds
- Excessive cost of guarantees
- Higher administrative costs and related risks resulting in lower contributions
- Mismanagement of the scheme by the employer or related parties resulting in leakages and /or effective returns being lower.
- Regulatory Non-compliance resulting in fines and penalties
- Any other valid point

[Max 8]

- ii) Key steps involved in modelling of the scheme
- The model should be fit for the purpose for which it is being used. It needs to be decided whether to go for a “off the shelf model” or build a model from scratch.
 - Decide whether to build a deterministic model, a stochastic model or combination. Both the models have advantages and disadvantages, the choice will depend on the availability of data and the expected output.
 - Define clear rules how the various parts interact with each other. E.g. how assets and liabilities interact with other, how the mortality rates vary etc.
 - The projected cashflows and their interactions to be modelled over their longer duration
 - Setting of assumptions and their correlations.
 - Considerable actuarial judgement will be needed in choosing the appropriate model, parameters and assumptions.
 - The scheme being very long term in nature will involve considerable effort in choosing appropriate models for interest rates, inflation and other economic factors.

- Be mindful of spurious accuracy. Strike a balance between frequency of cash flows and speed of prediction.
- The model should be able to calculate the liabilities, contributions, fund size etc. at least annually for the sponsor to make contributions and ensure viability of the scheme.
- The outputs from the model should be capable of independent verification for reasonableness.
- Build the model in such a way that the outputs are easy to interpret and communicate.
- Adequate documentation is required to be made while building the model.
- The model should be capable for development and refinement.
- Incorporate stress testing of the key assumptions and scenario testing based on the expected future economic scenarios.
- Any other valid point

[Max 8]

iii) The following cashflows need to be modelled

- Pension payments both guaranteed and discretionary if any
- Death benefits or any lumpsum payments to be made if any
- Commutation of pension and reinstatement of benefits.
- Transfer values of benefits in / out of the scheme
- Contributions made by the employer
- Investment income, dividends, interest income, maturity, redemption of the assets.
- Capital gains if any
- Administration / Investment / other expenses
- Taxes and other deductions
- Any other valid point

[Max 4]

iv) Assumptions used in the model and how the assumptions are set

Assumptions used in pension modelling

- Mortality: Working lives mortality, mortality post-retirement, spouses' mortality
- Retirement assumptions: Voluntary retirement, normal retirement, late retirements
- Withdrawal rates: Other than retirement and death
- New entrant rates and their average age
- Proportion married
- Average age of spouse, Average difference in age between employee and spouse
- Discount rate for discounting the liability
- Salary escalation rate: Inflationary increase, increase related to service, promotion etc
- Pension increases
- Price inflation, health inflation
- Expenses related to administration, investment etc.
- Investment returns, Yield curves
- Any other valid point

How to set assumptions

- DB pension being a long term liability, setting proper assumptions is very critical for the viability of the scheme.
- Assumptions are usually set based on the historical data though past performance may not necessarily be an indication of the future.
- Mortality assumptions may be set based on published tables with suitable adjustments based on the target population.
- Withdrawal and other rates may be set based on the experience of the company.

- Past experience of the company, published data from similar industries, national statistics, prevailing economic conditions, long term changes expected in the economic situation etc. may be considered while setting other demographic assumptions.
- The assumed mortality, withdrawal and retirement rates may be complied in multiple decrement tables to be used in the model.
- Increases in benefits may be linked to Inflation index and other published statistics
- Past and expected future salary increases either by industry, sector wise or country wide data may be used to set the salary escalation assumptions.
- Investment yield may be set based on the risk-free real yield plus expected inflation risk premium.
- Capital Asset Pricing Model (CAPM) may be an ideal method to set the investment yield bases on the invested assets.
- There may be policy statements published by central bank and other government bodies about the economic factors.
- Changes in legislations and related regulations need to be considered in monitoring the assumptions at regular intervals.

[Max 6]

v) CRO's concerns

- Being a FMCG major, the company may only have short-term liabilities in their books of accounts. The management may find it difficult to appreciate or manage the pension scheme liabilities which are very long term in nature. The company may consider to outsource the management to pension funds or hire experts to manage the same.
- Mortality risk: Depending on the promised benefits, higher pre- or post-retirement mortality may result in immediate lumpsum payout of death benefits resulting in loss of investment income. The management may consider insuring the death benefits.
- Longevity: Higher the longevity of beneficiaries than expected will result in longer duration of payouts. Purchasing annuity from a suitable provider may be one of the options to mitigate longevity risk.
- Risk of pandemics or one-off shocks may result in untimely payouts and higher benefit outgo thereby impacting the corpus and may create a liquidity crunch. The company may have to ensure that sufficient liquidity is always maintained at all times.
- Morbidity risk: Higher morbidity may result in higher ill-health retirements and related payouts.
- Investment / Market risk: Diversification of investment portfolios, investment in long term government / corporate bonds, equities etc. may help mitigate these risks
- Liquidity risk: Funds though sufficient in the long term may not be available when the benefit payments are required to be made. There may still be an issue of illiquidity for a funded scheme. But the risk will be more prominent for a funded scheme. Creating a separate fund for the scheme with proper choice of assets would mitigate the risk to a large extent.
- Expense risk: The risk that expense outgo is more than expected which is usually the case when the scheme is not managed properly or inexperienced hands do the administration. The company may consider outsourcing of the management with pre-agreed competitive expense ratios.
- Credit risk: Inherent to any investment portfolio. Diversification with portfolios, setting limits, due diligence etc. will help mitigate the risk.
- Model, parameter and data risk: The modelling of the scheme, the parameters considered and data may be inappropriate. This may result in inappropriate contributions or incorrect liabilities. Refining the model, benchmarking with other similar schemes, periodic actuarial valuations would help mitigate the risk to a large extent.
- Deterioration of employer covenant / Insolvency risk: Though the management is willing to make the contributions in time, profitability in business cycles may affect the payment

of contributions in time. One of the ways to mitigate this risk is to prevent reversion of any excess contribution to the sponsor by law.

- Contribution / Funding risk: The sponsor may not be able to pay the contributions in part or full or in time. The management may prefer a trust model with employee and employer representatives as trustees to safeguard the interests of the members.
- Risks of unviable guarantees: Make the scheme simple with guarantees if any to the bare minimum lest it turn out to be unviable in the future.
- Poor record keeping / administration. The scheme should have proper record keeping to avoid any litigations with stakeholders which may result in higher expenses and unnecessary leakages
- Conflicts of interest: There are likely to be different stakeholders with differing interests. The expectations of each of the stakeholder needs to be specified at the outset to avoid any conflicts thereafter.
- Ineffective relations with advisors: Accountants, scheme experts, actuaries, investment managers etc. are key to the viability of the scheme in the long run. Needless to say, effective relationship with them is more than a necessity.
- Any other valid point

[Max 10]

vi) Merits and demerits of funded scheme

- Funding increases the security of the scheme and also adds an element of credibility in the eyes of the employees.
- Employees may be willing to pay their share if any to the scheme if there is matching employer contribution.
- Lack of funding may affect the benefits payable if employer is facing financial difficulties
- Funding can be considered as easing of the burden of the employer in advance.
- Funding essentially helps to avoid raising contributions at potentially undesirable time in future.
- The contributions made to the pension fund may be tax-deductible for the employer.
- The employer may be willing to make higher contributions while making higher profits to compensate for any contributions to be made during tougher times.
- The regulation may specify the modes of investment of the fund resulting in non-optimum returns.
- The employer may always prefer to invest the funds in their business and expect to earn a higher return than that provided by other financial instruments.
- Employees don't have control over the funds or the investment policy of the fund.
- Pre-retirement investments may differ from those needed in the annuity phase.
- Even with proper funding, rates of inflation may be different from the investment returns, making the fund still inadequate to pay the benefits.
- Though funding eases the burden on the employer, it does not essentially absolve the employer of the responsibility to pay the promised benefits in time. The employer will still be required to make good for any shortfalls.

[Max 6]

vii) Disclosure requirements

- Disclosures are required as per the accounting standards or extant regulations. Public listed companies are required to disclose all the relevant information along with their annual accounts that may impact the financials in the short or long term.
- Disclosure of information about the significance of financials to an entity, nature and extent of risks involved, both qualitative and quantitative terms is one of the key requirements of IFRS and other accounting standards.
- Disclosure enhances the transparency of the scheme and the overall financials in general.

- Disclosure may help alert the trustees to any potential problems of the scheme. So that they are prepared to mitigate the risks.
- The company will be able to demonstrate the consistency in accounting treatment year on year and provide explanations for deviations if any.
- This is required for the shareholders to make informed decisions.
- The proposed pension scheme is a new employee benefit that is likely to have long term implications on the financials of the company.
- Proper disclosure and nod of the shareholders may be needed as this may necessitate raising of additional capital in the future.
- Reporting may be needed to demonstrate the solvency of the scheme and future contributions needed.
- Proper disclosure demonstrates the employer covenant and enhances the credibility of the scheme.
- Pension scheme is essentially a retention tool in the hands of the employer and disclosure of the scheme details may help attract appropriate human resources for the company.
- The disclosure is likely to include the benefits payable, contributions, risks involved, investment strategy, actuarial assumptions, calculated liabilities, surplus/deficit, membership movements and solvency of the scheme.
- Periodic disclosure helps set the expectations around the scheme for all stakeholders including the beneficiaries. This will help reduce any mis-representations and conflicts that may appear in the future.
- Disclosure may help members to exit the scheme (if an option exists) if the security / value of the benefits does not meet their expectations.
- Tax authorities will expect minimum prudence in the reserves of a company as it defers or reduces the tax payable. Hence it would be ideal to demonstrate the additional liabilities and the reasons for the same to avoid any future tax litigations.
- Any other valid point

[Max 8]

[50 Marks]
